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An Investigation of the Use of Network Analysis to Determine Evaluation Influence in Large, Multi-site NSF Evaluations



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American Evaluation Association Conference
Denver, Colorado
Thursday, November 6, 2008

Background:
The "*Beyond Evaluation Use*" Grant

- Grant to study use, influence, and involvement within NSF-funded programs.

- Research team (2 co-PIs and 4 graduate students) based at the University of Minnesota.

TARGETED CASE STUDIES

FOCUS QUESTIONS:

(1) What is the relationship between involvement in the multi-site evaluation and the use and influence of the evaluation on project leaders and evaluators?

(2) What factors appear to be related to the use and influence of the evaluation on the projects?

CONCEPTUAL FRAMES

- Evaluation use/ utilization and influence literature
- Participatory evaluation literature

METHODS

- Online survey (project leaders and evaluators)
- Interviews (project leaders and evaluators; NSF program officers)
- Review of archival documents, publications, and reports produced

INVOLVEMENT OF PROJECT LEADERS AND EVALUATORS, THEIR USE OF 4 NSF PROGRAM EVALUATIONS, AND THE EVALUATIONS' INFLUENCE

The Four Cases Studied

- Advanced Technological Education (ATE)
- Building Evaluation Capacity of STEM Projects - Math Science Partnership (MSP-RETA)
- Collaboratives for Excellence in Teacher Preparation (CETP)
- Local Systemic Change through Teacher Enhancement (LSC)

Examination of all cases concludes with evaluation efforts conducted through 2005

Approaches to examining influence on the field

- A citation analysis
- Surveys & follow-up interviews with NSF project principal investigators (PIs)
- Journal editor survey
- An online ‘influence on the field’ survey to detect presence of social network



Research Decisions

- How do we define the “field”?
  - A knowledgeable STEM audience that is likely to generate names or examples of influential evaluations, publications and evaluators.

- What do we mean by a social network?
  - ***Egocentric approach*** – placing the evaluator or program evaluation at the center of a model and showing how frequently “the field” found that person or program influential.

Survey Sample and Methods

- Randomly drawn sample
  - 350 Principal Investigators of active NSF grants
- Pre-pilot
- Think-alouds with experts
- Pilot of web-based survey
- Final online survey

Response Rate and Respondent Demographics

Response Rate **191/349** **54.7%**

Demographic Characteristics of Respondents		
Demographic Characteristic		***Percent***
Gender		
	Male	56.5
	Female	41.4
	No response	2.1
Number of NSF Grants the respondent was involved with at the time of the survey		
	1 NSF grant	34.0
	2-5 NSF grants	47.1
	> 5 NSF grants	17.3
	No response	1.6
Professional Background of Respondent		
	STEM Education/Researcher	52.4
	Scientist/Engineer/Mathematician	30.4
	Non-STEM Education/Researcher	6.8
	Other (Curator, Cultural Anthropologist, Economic Historian, Media Producer, Publisher)	6.3
	STEM Evaluator	1.6

Findings

- **Response Frame**: large scale, multi-site STEM education program evaluations
- 91 respondents (42.4%) were able to provide an evaluation, publication or evaluator BY NAME
- Of the 91 respondents, 48 listed an item in each category (evaluation, publication AND evaluator)

Findings

For what reason(s) do you think this evaluation has been influential? Check your top 2 choices:

Top Reasons – **Evaluation** Influential

It evaluated an important program	15
It was high quality work	15
It utilized sound evaluation/research	14
I was involved with it	12
It presented both qualitative and quantitative findings	11

Findings

For what reason(s) do you think this evaluation publication has been influential? Check your top 2 choices:

Top Reasons – **Publication** Influential

It presented both qualitative and quantitative	14
It provided important recommendations	13
It was high quality work	12
It has had far-reaching influence on both policy and practice	10
It presented sound evaluation/research methodology	9

Findings

This evaluator has been influential because he/she (check your top 2 choices):

Top Reasons – **Evaluator** Influential

Has demonstrated a deep understanding of STEM education	17
Has utilized sound evaluation/research methodology	11
Contributed significantly to the knowledge base of evaluation	11
Has had considerable STEM evaluation experience	10

Findings

What influence did the four programs being studied have on the field?

Named an **evaluation** related to one of the four programs	15 of 64
MSP	6
ATE	4
LSC	4
CETP	1

Named a **publication** related to one of the four programs	13 of 57
ATE	4
LSC	4
MSP	4
CETP	1

Findings

What influence did the evaluators associated with these four programs being studied have on the field?

Most named evaluators	n	Percent of total respondents
An evaluator associated with LSC	7	3.7%
An evaluator associated with CETP	3	1.6%
An evaluator associated with American Association for the Advancement of Science	2	1%
An evaluator associated with Inverness Research Associates	2	1%
An evaluator associated with University of Colorado	2	1%
An evaluator associated with Grinnell College	2	1%

70 respondents were able to name an evaluator by name.

Findings

Most named organization affiliation of evaluators	n	Percent of total respondents
Horizon Research Inc.	7	3.7%
American Association for the Advancement of Science	4	2.1%
University of Minnesota	3	1.6%
University of Wisconsin Center for Education Research	2	1%
University of Washington	2	1%
University of Colorado	2	1%
SRI International	2	1%
Michigan State University	2	1%
Inverness Research Associates	2	1%
Grinnell College	2	1%

Discussion

- Strengths
  - Systematic process for developing instrument and sampling plan
  - High response rate
- Speculations
  - Sampling plan and methods may not have been sensitive enough to detect a social network
  - NSF evaluations may only inform a limited, specialized community of practice
- Implications for future research

Contact Information

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